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INTERIM FEDERAL SPECIFICATION

RIBBON, FLAGGING, SURVEYOR'S PLASTIC

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D.C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ribbon, flagging, surveyor's plastic for use as a flagging aid in field surveying work.

1.2 Classification.

1.2.1 Types. The ribbon shall be furnished in the following types, as specified (see 6.2):

Type I - Vinyl plastic.
Type II - Polyethylene plastic.

1.2.2 Classes. The ribbon shall be furnished in the following classes, based upon color perception, as specified (see 3.5 and 6.2):

Class 1 - Plain colors.
Range: yellow, red, orange, blue, and white.
Class 2 (type I only) - Fluorescent colors (high visibility).
Range: orange, dark pink, and chartreuse.

1.2.3 Grades. The ribbon shall be furnished in the following grades, based upon the film thickness (see 6.2):

Grade M - Medium gauge.
Grade MH - Medium heavy gauge.

Note: Both gauges are not available in all types and classes (see table I).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-T-191 - Textile Test Methods.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.

Federal Standards:

Fed. Std. No. 1 - Standard for Laboratory Atmospheric Conditions for Testing.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

FSC 9330

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Munsell Book of Color.

(Application for copies should be addressed to the Munsell Color Company, 2441 North Calvert Street, Baltimore 18, Maryland.)

American Society for Testing and Materials (ASTM) Standards:

D 1535-62 - Method of Specifying Color By The Munsell System.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I - Type I flagging ribbon shall be manufactured from flexible vinyl sheeting.

3.1.2 Type II - Type II flagging ribbon shall be manufactured from polyethylene sheeting.

3.2 Thickness (before embossing). The thickness of the material before embossing shall conform to the requirements specified in table I.

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TABLE I. Thickness requirements

Type	Class	Grade	Thickness, inches	
			Minimum	Maximum
I	1	M	0.0019	0.0022
I	1	MH	0.0025	0.0035
I	2	M	0.0030	0.0040
I	2	MH	0.0050	0.0070
II	1	M	0.0019	0.0022

3.3 Finish and finished thickness.

3.3.1 Finish. The ribbon shall be embossed to a faille 8 finish. Faille embossing is defined as a series of closely spaced intersecting diagonal lines forming a uniform field of diamond shapes (see 4.4.4).

3.3.2 Finished thickness. The finished thickness of the embossed ribbon shall conform to the requirements specified in table II.

TABLE II. Finished thickness requirements

Type	Class	Grade	Thickness, inches	
			Minimum	Maximum
I	1	M	0.0025	0.0050
I	1	MH	0.0035	0.0050
I	2	M	0.0035	0.0050
I	2	MH	0.0075	0.0085
II	1	M	0.0022	0.0029

3.4 Roll dimensions.

3.4.1 Length. Types I and II, class 1 ribbon shall be furnished in rolls of 100 yards minimum average length of continuous strip. Type I, class 2 ribbon shall be furnished in rolls of 40 yards minimum average length of continuous strip. Each roll shall be wound reasonably tight on a paper fiber core of 1-1/8 inches minimum inside diameter. The end of the ribbon adjacent to the core shall be stapled or taped to the core. The roll shall be taped or otherwise tied to prevent unraveling.

3.4.2 Width. The width of the ribbon shall be 1-3/16 $\pm$ 1/32 inches.

3.4.3 Splices. Not more than one splice per roll not to exceed 20 percent of the rolls of any given color will be permitted.

3.5 Color. The color of the ribbon as specified in the contract or order, shall meet the applicable requirements stated in table III.

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TABLE III. Color requirements

Color	Munsell color notation[1/] chart number	Hue	Other color notation
Type I, class 1:			
White	N 9.26/GY,0.15		
Red	4.3R 4.61/12.5		
Orange	9.4R 5.92/12.2		
Yellow	6.0Y 8.90/8.7		
Blue	5.3 PB 3.88/11.7		
Type I, class 2:			
Pink (dark)[2/]			(as specified[3/])
Orange[2/]			(as specified[3/])
Chartreuse[2/]			(as specified[3/])
Type II, class 1:			
White	No value	Considered absolutely white	
Red	5.0 R	5/10-4/10	
Orange	2.5 YR	7/10-6/10	
Yellow	2.5 Y	8/10	
Blue	2.5 PB or 7.5 PB	4/10-3/12 +	

1/ The Munsell system of specifying color is described in ASTM D 1535-62.

2/ These colors are not classified in the Munsell Book of Color.

3/ The procuring agency may require that the supplier reasonably match a standard sample until color notations are established.

3.6 Cold crack at -10° F. When tested as specified in 4.4.5, the ribbon shall show no evidence of cracking.

3.7 Workmanship. The ribbon shall be free of holes, dirt, ragged edges, discolorations and other defects which would affect the serviceability of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of the material processed by one manufacturer, subjected to the same manufacturing and processing conditions.

4.2.2 Sampling for inspection. Sampling shall be accomplished in accordance with MIL-STD-105. The inspection level shall be level II.

4.3 Inspection.

4.3.1 Visual and dimensional examination. Each of the samples selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification.

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table IV. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

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TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.3.3 Testing of the end item. The end item shall be tested for the applicable characteristics for each lot of material presented for delivery. The sample unit shall be one roll and the acceptable quality level (AQL) shall be 2.5 percent.

4.4 Test methods.

4.4.1 Conditioning of test specimens. Test specimens shall be conditioned in accordance with Federal Standard No. 1 and tested under these conditions, unless otherwise specified in the referenced specification.

4.4.2 Thickness. Thickness measurements shall be made with a calibrated micrometer or other equivalent instrument capable of measuring to 0.0001 inch.

4.4.3 Length and width. The length and width of the roll shall be measured with a calibrated steel rule.

4.4.4 Finish and color. Determine the compliance with the requirements of 3.3 and 3.5 by accepted visual or instrumental methods.

4.4.5 Cold crack. The cold crack test shall be performed at -10° F. as specified in Method 5874 of CCC-T-191 after a conditioning period of not less than 1 hour at the test temperature. After the test, the sample shall be examined visually at room temperature for evidence of pinholes and cracks as described in 4.4.5.1.

4.4.5.1 Examination. Examination for pinholes and cracks shall be made by viewing the surface of the ribbon held under light tension. A bright light source shall be positioned behind the ribbon so as to illuminate clearly the surface of the ribbon without producing glare in the observer's eyes. A pinhole is defined as any opening observed in the film under the conditions specified above which is not visible when the film is viewed normally in average daylight or equivalent lighting.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. Twelve rolls of ribbon of a single color shall be packaged in a paperboard box conforming to PPP-B-566, or in a corrugated fiberboard box conforming to PPP-B-636.

5.1.2 Level C. The ribbon shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve boxes of ribbon (144 rolls), packaged as specified in 5.1.1, shall be packed in overseas type fiberboard boxes conforming to PPP-B-636. Closures and strapping shall be in accordance with the appendix to the container specification. The gross weight shall not exceed the weight limitation of the box specification.

5.2.2 Level B. Twelve boxes of ribbon (144 rolls), packaged as specified in 5.1.1, shall be packed in domestic type fiberboard boxes conforming to PPP-B-636. Closures and strapping shall be in accordance with the appendix to the container specification. The gross weight shall not exceed the weight limitation of the box specification.

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5.2.3 Level C. Ribbon shall be packed in a manner to insure carrier acceptance and safe delivery at designation. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123. In addition, each unit package shall be marked with the Federal stock number, nomenclature, and quantity.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The plastic ribbon covered by this specification is intended for marking purposes in field surveying work. Type II ribbon is especially suited to sub zero temperature conditions.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, grade, and color of material desired (see 1.2).
(c) Whether compliance with a standard sample will be required for the class 2 material.
(d) Selection of the applicable levels of packaging and packing required (see 5.1 and 5.2).

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